

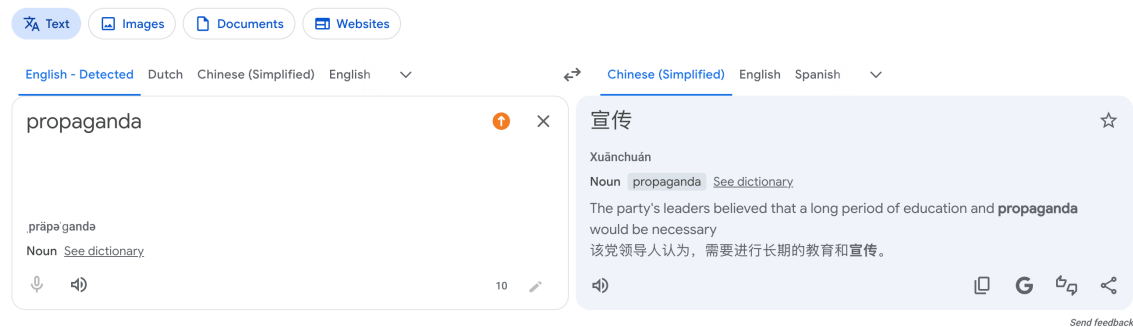
Grounding Cross-Lingual Conceptual Equivalence in Human Associations

Central argument:

Language- and culture-specific word meanings are often not fully captured by dictionary translations. These hidden differences can cause misunderstandings in cross-cultural communication because the translated word may look similar while changing the speaker's attitude, evaluation, or ideological implication.

For example, In English, “propaganda” is usually negative, suggesting dishonest, manipulative communication. In Chinese, “propaganda” is usually translated into the Chinese *xuanchuan* 宣传, which is one word much more neutral and sometimes even positive term, referring to the mass spread of information. This translation was used in the English name of “中共中央宣传部”, “The Chinese Communist Party Central Committee **Propaganda** Department”, which is the Chinese Communist Party's central department for public communication. In 1998, the Party noticed the reactions that the word propaganda elicited and changed it to “Publicity”.

Google Translate



Such hidden meanings can create misunderstandings in cross-cultural communication among people. For a second-language speaker who has not acquired the social and cultural experience behind these words, the difference is difficult to detect: the dictionary translation may look correct, while the attitude and ideological positioning have shifted. Likewise, they may also remain hidden in NLP models trained mainly on text, which lack the fully embodied environmental and cultural experience.

Meanwhile, modern multilingual NLP needs reliable representations of concepts across languages. As large language models in NLP are increasingly used by global users across languages and cultures, it becomes important to evaluate whether they can accurately understand the hidden meanings behind, compare subtle concepts in different languages, and generate response that can effectively communicate with global users. █

Recent NLP work has developed cross-lingual datasets and benchmarks for evaluating how models represent word meanings across languages. **Existing word-level evaluations often treat meaning comparison as a problem of translation equivalence or scalar similarity.** Specifically, cross-lingual word-level resources and evaluation methods often operationalize meaning alignment through either translation links (whether two words are translatable) or scalar similarity scores (indicate how close two words are). Both are useful but lack explainability on how and why they are more similar or less similar. Two words across languages may share a core concept while differing in specificity, affective tone, social

implications, etc. Reducing cross-lingual semantic equivalence to a single similarity score collapses this internal structure and multidimensional nature of meanings.

As a result, it obscures the very distinctions that matter for understanding human meaning: where concepts overlap, where they diverge, and how such divergences shape cross-cultural miscommunication. On the technological side, it constrains the development of more aligned language technologies that can translate, explain, and adapt meanings in ways that are genuinely aligned with human communication.

We therefore need a new framework that can both measure and explain conceptual equivalence across languages. In this study, we propose a graded, interpretable, and context-grounded framework for cross-lingual lexical semantic equivalence. This framework estimates not only how close two words are, but also which meanings are shared, which meanings differ, and what evidence supports the judgment.

Let $x^s \in V_s$ and $x^t \in V_t$ be two words from languages s and t . **We define cross-lingual lexical semantic equivalence as the task of estimating $\text{Eq}(x^s, x^t)$, the degree to which x_s and x_t are conceptually equivalent, together with interpretable evidence for this estimate.** To ground this judgment in human meaning, we represent each word by its association set: $\mathcal{A}_s(x^s) = \{(a_i^s, x_i^s)\}_{i=1}^m$ and $\mathcal{A}_t(x^t) = \{(a_j^t, x_j^t)\}_{j=1}^n$. Here, a denotes an associated word and w denotes its association strength. Our hypothesis is that greater alignment between $\mathcal{A}_s(x^s)$ and $\mathcal{A}_t(x^t)$ indicates stronger conceptual equivalence between x^s and x^t , while weak alignment indicates partial or limited equivalence.

The association set is derived from free word association, a psychological behavioural task in which participants respond to a cue word with the words that first come to mind. For example, given a word *smile*, native speakers might respond with *happy*, *teeth*, *camera*. *We use word associations as cognitively grounded, observable traces of how humans represent word meaning.* Prior work shows that large-scale word association norms, including the USF norms [2] and English SWOW [3], can predict human behaviour in lexical decision, semantic categorization, and similarity judgment tasks, suggesting that associations capture psychologically meaningful lexical knowledge. Multilingual SWOW resources [4,5,6,7] and its follow-up work further show that associations provide, structured representations of commonsense knowledge [11, 12], as well as visual, affective [8], cultural [10], and value-laden [9-10] dimensions of meaning that may not be fully recoverable from text corpus-based embeddings alone. We therefore treat associations as cognitively grounded and interpretable evidence for how native speakers mentally represent word meanings.

This approach allows cross-lingual conceptual equivalence to be evaluated not only as a score, but as an interpretable alignment between meaning structures. The framework can reveal what is shared, what is language-specific, and why two words should be judged as equivalent, partially equivalent, or different.

The contribution of this work is therefore to define cross-lingual word semantic equivalence as a graded, explainable, and human-grounded problem, and to provide resources and methods to (1) automatically quantify concept equivalence and (2) evaluate whether multilingual models understand word meanings across languages.

We decompose the goal into three tasks.

1. **Task 1: Define.** We formalize cross-lingual lexical semantic equivalence as a graded relation between two words/concepts across languages. This includes situating conceptual equivalence with other related concepts: translation, semantic similarity, relatedness, and broad topical association. Given a pair of cross-lingual words, we consider four levels of equivalence between them:

Cross-Lingual Semantic Alignment

- |-- Direct translation equivalence (1.0)
- |-- Near-equivalence / strong semantic similarity (0.8)
- |-- Partial semantic overlap (0.5)
- |-- No semantic overlap (0.0)

Although graded judgments have been widely used in prior word similarity benchmarks [13–15], our definition of cross-lingual lexical semantic equivalence is deliberately stricter. We distinguish equivalence from thematic relatedness, contextual association, and broad topical co-occurrence. For example, in prior similarity datasets, word pairs such as *house* and *window* may receive a non-zero similarity score (e.g., 2 out of 4, slightly similar) because they are situationally related and share a common topic or domain. Under our framework, however, this pair would receive a score of 0, because *house* and *window* denote different concepts and do not share the same core conceptual meaning. This distinction is essential for cross-lingual evaluation: when comparing words across languages, we need to estimate whether they express the same or overlapping meaning, not merely whether they appear in related situations or topics. By excluding thematic association from equivalence judgments, our framework provides a more precise account of conceptual overlap and avoids overestimating semantic alignment between words that are related but not equivalent.

Formally, we define this task as a function g that takes two words in and return an equivalence rating:

$$\text{Eq}_{\text{direct}}(x^s, x^t) = g(x^s, x^t) \in [1.0, 0.8, 0.5, 0]$$

where $g(x^s, x^t)$ assigns one of four equivalence scores: 1.0 for direct translation equivalence, 0.8 for near-equivalence or strong semantic similarity, 0.5 for partial semantic overlap, and 0.0 for no semantic overlap.

The same function g can be applied the (x^s, x^t) pair. It can also be applied directly to any pair of cross-lingual words to estimate their conceptual equivalence.

2. **Task 2: Represent.** We represent each cue word through an association-based meaning context. Rather than treating a word as an isolated lexical item, we model its meaning through the list of mental associations it evokes among native speakers,

capturing the mental representations connected to the word.

$$\begin{aligned}\mathcal{A}_s(x^s) &= \{(a_i^s, x^s)\}_{i=1}^m, \\ \mathcal{A}_t(x^t) &= \{(a_i^t, x^t)\}_{i=1}^n\end{aligned}$$

We retain the top-percentile responses to focus on stable associations and reduce noise. [CL: how to choose the association subset is still not crystal clear.]

3. **Task 3: Align.** We directly compare and align the association contexts of two cross-lingual words using a cross-lingual association matrix M and quantify their degree of semantic overlap using an aggregation function $f(\cdot)$. This produces a graded equivalence score and supports qualitative interpretation of where the two meanings align or diverge.

$$\begin{aligned}M_{ij} &= g(a_i^s, a_j^t), \quad M_{ij} \in \{1.0, 0.8, 0.5, 0.0\}, \quad M \in \mathbb{R}^{m \times n}, \\ \text{Eq}_{\text{assoc}}(x^s, x^t) &= f(M) \in [0, 1], \\ f(M) &= \frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n M_{ij}\end{aligned}$$

The function (f) aggregates the alignment matrix (M) into an overall equivalence score for the word pair, and can be instantiated in different ways, such as mean, maximum, density, or frequency-weighted aggregation.

By comparing the correlation between the direct equivalence mapping $\text{Eq}_{\text{direct}}(x^s, x^t)$ and the association-based equivalence $\text{Eq}_{\text{assoc}}(x^s, x^t)$, we can estimate how much associative context contributes to lexical meaning alignment. Meanwhile, the matrix M provides evidence and explainability of how and where the equivalence is mapped.

Through this framework, the project contributes a more interpretable and human-centered account of cross-lingual word meaning measurement. It shifts the problem from simply matching words across languages to understanding how meanings travel, transform, and diverge across linguistic and cultural communities.

Ultimately, this work argues that cross-lingual lexical meaning should not be treated as a fixed one-to-one mapping. It should be modelled as a graded, explainable, and context situated relation between concepts.

What is new in this work?

This work contributes a new framework, resource, and empirical analysis for explainable cross-lingual lexical semantic equivalence. We make four contributions:

1. **A graded semantic alignment framework for lexical equivalence.** Cross-lingual word pairs are positioned along a continuum from direct translation equivalence to near-equivalence, partial semantic overlap, and non-equivalence. This provides a clearer account of how translation, semantic similarity, and conceptual

overlap relate to one another.

2. **An interpretable account of equivalence based on associative meaning**, making it possible to explain which aspects of meaning are shared or language-specific. Word meanings are represented through human association sets, allowing equivalence judgments to be explained through shared, partially shared, and language-specific associations rather than only through a scalar score.
3. **A multilingual resource linking direct equivalence and association-based meaning**.
The resource covers five languages: English, Chinese, Dutch, German, and Spanish, pairing direct cross-lingual equivalence judgments with association-based semantic contexts. This supports both scoring and explanation of cross-lingual meaning alignment.
4. **Empirical evidence on cross-lingual meaning divergence**.
[placeholder: It offers empirical evidence on how lexical meanings align and diverge across languages, including cases where apparently translatable words differ in associative, cultural, affective, or evaluative meaning.]

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Version 1 (Deprecated)

The rationale of this need for this work:

- (1) Recent advances in cross-lingual and multilingual NLP have made it possible for models to process and generate text across many languages. Large language models are trained on large-scale multilingual data and are expected to encode, compare, and transfer meanings across languages.
- (2) These models are increasingly used not only for translation, but also for multilingual retrieval, cross-lingual reasoning, and generation. In these settings, the system must not only understand meanings expressed in different languages, but also generate outputs that preserve, compare, and adapt those meanings across linguistic and cultural contexts.
- (3) This makes cross-lingual meaning representation a central problem. If a model cannot correctly represent how meanings correspond across languages, higher-level tasks that depend on those meanings may also become unreliable.

(4) A fundamental part of this problem occurs at the lexical level. High-level semantic understanding relies on lower-level meaning representations, including how individual words and concepts are encoded. If word-level meanings are misaligned across languages, then sentence-level, discourse-level, and task-level understanding can also be affected.

(5) As AI systems become more widely used across multilingual and multicultural contexts, it is no longer sufficient for them to produce broadly correct translations or approximate semantic matches. They also need to explain subtle differences between words, concepts, and cultural meanings across languages.

(6) However, existing cross-lingual word-level resources and evaluation methods often operationalize meaning alignment through either translation links or scalar similarity scores. Translation links indicate that two words can correspond in use, while similarity scores indicate how close two words are. Both are useful, but neither explains why two words are aligned, in what sense they are similar, or where their meanings diverge.

(7) This is a major limitation because word meaning is not a single undifferentiated quantity. A word's meaning may involve conceptual scope, emotional tone, cultural grounding, social usage, pragmatic implication, and associative meaning. Two words may be similar in one aspect but different in another.

(8) Existing similarity ratings also often hide the evidence behind the judgment. Annotators may provide a similarity score, but the meaning structure that led to the score remains hidden and untraceable. As a result, **the semantic construct being evaluated is implicit, difficult to inspect, and may vary from word pair to word pair.**

(9) Therefore, there is a gap in current evaluation: we lack datasets and methods that can evaluate models' understanding of cross-lingual word meaning in a way that is both graded and interpretable.

(10) This gap motivates us to move beyond treating cross-lingual word alignment as either translation correspondence or an opaque similarity score.

(12) We define the problem as **cross-lingual lexical semantic equivalence**: given a word (x_s) in language (s) and a word (x_t) in language (t), the goal is to determine the degree to which the two words express the same or comparable meaning.

(13) The goal is not only to ask how a word is semantically equivalent to another word, but to determine how far their meanings overlap, where they diverge, and what kinds of meaning differences matter.

(14) **To address this, we propose to ground cross-lingual semantic equivalence in human associations.** Instead of treating each word as an isolated lexical item, we represent it through the list of its mental associations it evokes for speakers.

(15) If two words are semantically equivalent, their association-based meaning contexts should also align. If their association spaces diverge, this provides evidence that the two words are only partially equivalent or non-equivalent.

(16) This allows cross-lingual semantic equivalence to be evaluated not only as a score, but as an interpretable alignment between meaning structures. The framework can reveal what is shared, what is language-specific, and why two words should be judged as equivalent, partially equivalent, or different.

(17) The contribution of this work is therefore to define cross-lingual word semantic equivalence as a graded, explainable, and human-grounded problem, and to provide resources and methods for evaluating whether multilingual models understand word meanings across languages.

(18) We decompose the project into three tasks.

1. **Task 1: Define.** We formalize cross-lingual lexical semantic equivalence as a graded relation between two words/concepts across languages. This includes distinguishing equivalence from translation, semantic similarity, relatedness, and broad topical association. It is a general semantic equivalence framework that integrates translation equivalence and conceptual equivalence. We consider four levels of equivalence:

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- |-- Partial semantic overlap (0.5)
- |-- No semantic overlap (0.0)

2. **Task 2: Represent.** We represent each word through an association-based meaning context. Rather than treating a word as an isolated lexical item, we model its meaning through the list of mental associations it evokes among native speakers, capturing the mental representations connected to the word.
3. **Task 3: Align.** We compare the association contexts of two cross-lingual words and quantify their degree of semantic overlap. This produces a graded equivalence score and supports qualitative interpretation of where the two meanings align or diverge.

Formally, it can be defined by:

$$\begin{aligned}
 \mathcal{R}_x &= r_1^x, \dots, r_m^x, \\
 \mathcal{R}_y &= r_1^y, \dots, r_n^y \\
 M_{ij} &= \text{align}(r_i^x, r_j^y), \quad M_{ij} \in \{1.0, 0.8, 0.5, 0.0\}, M \in \mathbb{R}^{m \times n}, \\
 \text{Eq}(w_x, w_y) &= f(M) \in [0, 1], \\
 f(M) &= \frac{1}{mn} \sum_{i,j} M_{ij}.
 \end{aligned}$$

where $\text{align}(r_i^x, r_j^y)$ assigns one of four semantic alignment scores: 1.0 for direct translation equivalence, 0.8 for near-equivalence or strong semantic similarity, 0.5 for partial semantic overlap, and 0.0 for no semantic overlap. The function (f) aggregates the alignment matrix (M) into an overall equivalence score for the word pair, and can be instantiated in different ways, such as mean, maximum, density, or frequency-weighted aggregation.

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Ultimately, this work argues that cross-lingual lexical meaning should not be treated as a fixed one-to-one mapping. It should be modelled as a graded, explainable, and context situated relation between concepts.